Energy & efficiency

Higher · Engineering contexts & challenges · Worksheet

ENGINEERING WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ describe energy conversions in a system and draw an energy audit
- ☐ calculate kinetic, potential, electrical and heat energy
- ☐ calculate power, work done and efficiency
- ☐ explain ways of reducing energy losses
- ☐ solve multi-stage energy problems involving efficiency and losses at each stage

I DO — Worked example

Read how a full-mark answer is built

The lift motor uses 95 kJ of electrical energy to raise the car (Ep = 76 440 J).

1 Calculate the efficiency of the lift. (2 marks)

Full-mark answer each ✓ is one mark

η = useful energy out ÷ energy in ✓ = 76 440 ÷ 95 000 = 0.80 or 80% ✓.

Why answers lose marks

"η = 95 000 ÷ 76 440 = 1.24"

0 marks: values upside down. Efficiency can't be more than 1 (100%).

WE DO — Guided practice

Use the starters and word bank to help you

A lift raises a 650 kg car through 12 m.

2 Calculate the potential energy gained by the car. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Kinetic energy depends on...
☐ Height ☐ Mass and velocity ☐ Current and time ☐ Temperature

b) The unit of power is...
☐ Joule ☐ Newton ☐ Watt ☐ Volt

c) Unwanted energy in a gearbox is mainly lost as...
☐ Light ☐ Heat and sound ☐ Potential energy ☐ Chemical energy

d) A 24 V, 0.42 A device runs for 7.5 hours. Which time value must be used?
☐ 7.5 ☐ 450 ☐ 27 000 ☐ 0.125

4 Key terms: write the letter of the matching description next to each term.

1. **Work done**	____	**A** $P = E \div t$
2. **Potential energy**	____	**B** $E = \frac{1}{2} m v^2$
3. **Kinetic energy**	____	**C** $E = m \times g \times h$ ($g = 9.8$)
4. **Electrical energy**	____	**D** $E = V \times I \times t$ (t in seconds)
5. **Heat energy**	____	**E** $E = c \times m \times \Delta T$
6. **Power**	____	**F** $E = F \times d$

An e-bike and rider have a total mass of 110 kg and travel at 7 m/s.

5 Calculate the kinetic energy. (2 marks)

A kettle element is rated at 2.2 kW.

6 Calculate the energy used in 3 minutes. (2 marks)

1.5 kg of water is heated from 15 °C to 100 °C. Specific heat capacity of water = 4180 J/kg °C.

7 Calculate the heat energy needed. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?